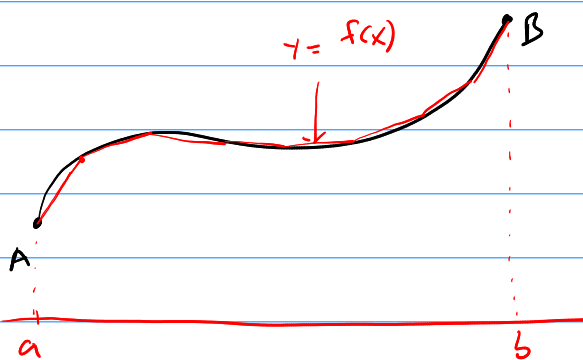


Chapter 5: Applications of Definite Integrals

Section 5.4: Length of a plane curve

Curves: $\begin{cases} \nearrow \text{2-D curve (plane curve)} \\ \searrow \text{3-D curve} \end{cases}$



Def : If $y = f(x)$ is a "smooth" curve on the interval $[a, b]$

then the arc length L of this curve over

$[a, b]$ is defined as

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} \, dx$$

or

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \, dx$$

Note: If the curve is present by $x = h(y)$

then the length is defined as

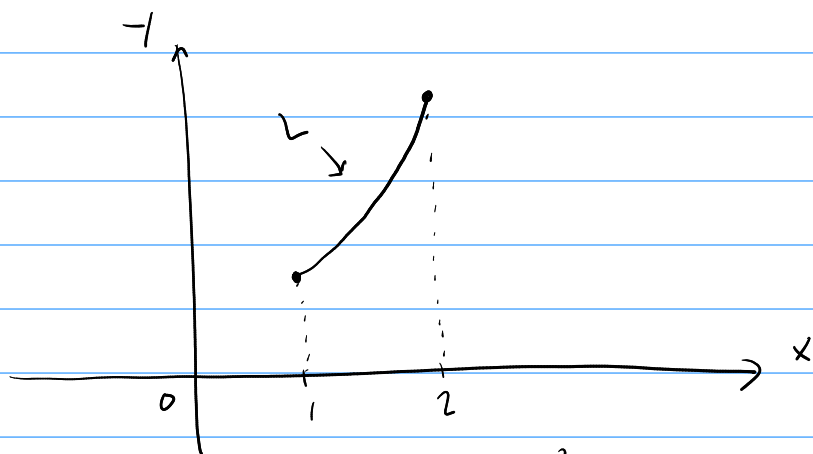
$$L = \int_a^b \sqrt{1 + h'(y)} \, dy \quad \text{OR}$$

$$L = \int_a^b \sqrt{1 + \left(\frac{dx}{dy}\right)^2} \, dy$$

Example :

Find the length of the curve $y = x^{3/2}$ from

point $(1, 1)$ to point $(2, 2\sqrt{2})$



we have:

$$L = \int_1^2 \sqrt{1 + [f'(x)]^2} \, dx$$

$$f(x) = x^{3/2} \Rightarrow f'(x) = \frac{3}{2} x^{3/2 - 1}$$

$$\Rightarrow f'(x) = \frac{3}{2} x^{1/2} = \frac{3}{2} \cdot \sqrt{x}$$

$$\Rightarrow L = \int_1^2 \sqrt{1 + [f'(x)]^2} \, dx$$

$$= \int_1^2 \sqrt{1 + \left(\frac{3}{2} \sqrt{x}\right)^2} \, dx$$

$$= \int_1^2 \sqrt{1 + \frac{9}{4} x} \, dx$$

We will use substitution method.

$$\text{let } u = 1 + \frac{9}{4} x$$

x	1	2
u	$\frac{13}{4}$	$\frac{11}{2}$

$$\left[1 + \frac{9}{4}\right]$$

$$\left[1 + \frac{9}{4} \cdot 2\right]$$

$$\text{we have: } du = \left(1 + \frac{9}{4} x\right)' \cdot dx$$

$$\Rightarrow du = \frac{9}{4} \cdot dx$$

$$\Rightarrow dx = \frac{4}{9} du$$

$$L = \int_1^2 \sqrt{1 + \frac{9}{4}x} \, dx$$

$$= \int_{13/4}^{9/2} \sqrt{u} \cdot \frac{4}{9} du$$

$$\int u^n du = \frac{u^{n+1}}{n+1} + C$$

$$= \frac{4}{9} \int_{13/4}^{9/2} u^{1/2} \, du$$

$$= \frac{4}{9} \left[\frac{u^{1/2+1}}{1/2+1} \right]_{13/4}^{9/2}$$

$$= \frac{4}{9} \left[\frac{u^{3/2}}{3/2} \right]_{13/4}^{9/2}$$

$$= \frac{8}{27} \left[\left(\frac{9}{2} \right)^{3/2} - \left(\frac{13}{4} \right)^{3/2} \right] \approx 2.09$$

Find the exact arc length of the curve over the interval.

Find the exact arc length of the curve over the interval.

$$y = x^{2/3} \text{ from } x = 1 \text{ to } x = 8$$